Medical Tourism and Clinic Clustering

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Health Inequality and Medical Tourism

- Medical tourism consists of temporarily visiting a country with the main objective of attaining medical care.
- Key Reasons
  - Lower Costs
  - Shorter Wait Times
  - Better Healthcare
  - Legal differences
- Medical tourism has contributed to the economic growth and specialization of certain cities, but has also brought unintended consequences

- Los Algodones, Baja California—also known as “Molar City.” [Clement et al., 2024]
  - The city has an exceptionally high concentration of dental care clinics serving foreign patients.
  - Local marginalized populations may be adversely affected through a crowding-out effect, as resources and services cater primarily to medical tourists.
- Insulin and Fertility Treatment
  - Procedures and medicines that are expensive in the U.S. can cost a fraction in Mexico.
  - Tijuana, Monterrey, and Juarez are the biggest cities for this type of medical tourism.
  - IVF and Insulin for example can be cheaper.
  - IVF (In vitro fertilization) which can be around $18,000 in the us can cost $7,500 in Mexico
  - Insulin can be purchased for reduced prices like for example $16 vs $110 for an insulin pen

Question and Data Source

Question

- Is there any evidence of clustering of medical units on U.S./Mexico border towns?

Data

- The data was collected for 2025 September.
- Data on medical units comes from the Mexican Secretary of Health; it is the universe of codes given to all medical units in Mexico, both private and public. The data contains coordinates for each establishment, as well as several indicators for rural/urban and other types of care.
- Data on population per municipalities and the shape file for all of Mexico comes from the Mexican National Institute for Statistics and Geography.

Summary Statistics

Table 1: Summary Statistics for Population and Clinic Measures

Variable	Min	Q1	Median	Mean	Q3	Max
Population	349	3,098	9,230	89,487	44,852	2,151,740
Number of Clinics	1.00	5.00	10.50	33.32	28.00	635.00
Expected Clinics	0.13	1.15	3.44	33.32	16.70	801.11
Clinics per 10,000	0.75	4.19	9.87	14.05	19.19	80.65

Exploratory Analysis

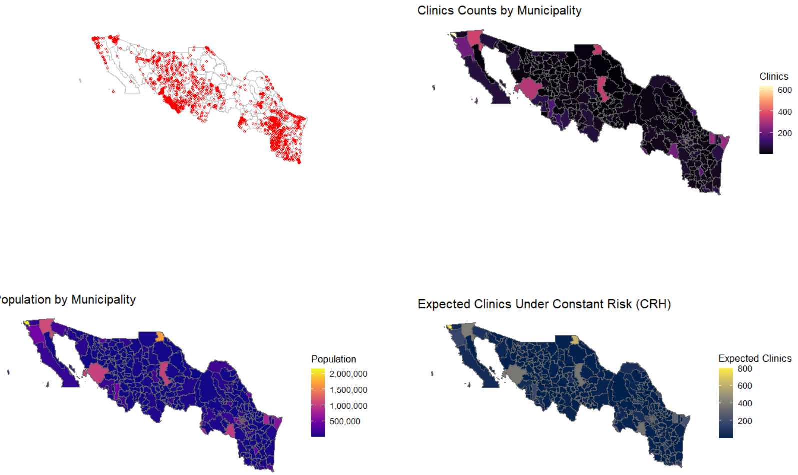


Figure 1: Clinic Distribution, Population, and Expected Clinics Under Constant Risk

Besag-Newell Test

- Local Test

Ho The most compact window regarding population with at least c* hospitals on the border U.S. Mexico is not significantly more compact than what is expected under the constant risk hypothesis

Ha The most compact window regarding population with at least c* hospitals on the border U.S. Mexico is significantly more compact than what is expected under the constant risk hypothesis

Besag–Newell Method

Threshold c^*	Cases	Population	p-value
10.5	11	1,364	1.98×10^{-10}
28	28	5,091	3.15×10^{-23}
33.32	39	7,806	1.34×10^{-23}
68.3	71	16,118	9.06×10^{-47}

Table 2: Besag–Newell cluster results for Northern Mexico across multiple c^* thresholds.

- The way I picked the c* was by getting the mean and the following quantiles 50th, 75th, 90th,

Besag–Newell Method Maps

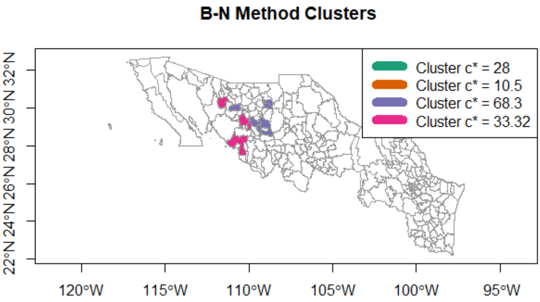


Figure 2: Besag-Newell Clusters

CEEP Test

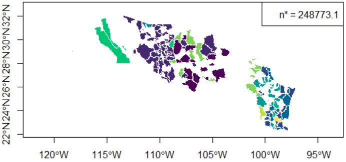
- Local Test

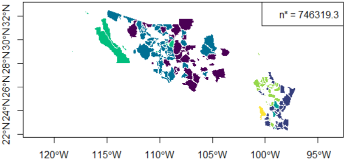
Ho There is no window with n* persons that has significantly more clinics in the U.S. Mexico border states than what is expected under the CRH

Ha There is at least one window with n* persons that has significantly more clinics in the U.S. Mexico border states than what is expected under the CRH

- The way I picked the n* was by getting clusters that contain 1% 3% and 5% of the total population. Since this helps us look at very small local clusters to larger regional concentrations of populations.
- We see that at these n* values the test stays robust and consistent apart from some major metropolitan areas like Tijuana.

CEEP Test Maps





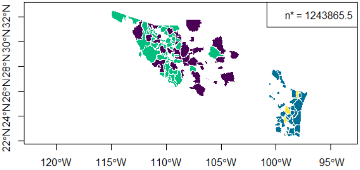


Figure 3: CEPP Clusters at Multiple Population Scales

Scan Test (Kulldorff)

- Local Test

Ho The most likely cluster of clinics is consistent with what is expected under the constant risk hypothesis, regarding the local rate of clinics compared outside of the cluster

Ha The most likely cluster of clinics is consistent with what is more extreme than what is expected under the constant risk hypothesis, regarding the local rate of clinics compared outside of the cluster

Scan Test (Kulldorff) Map

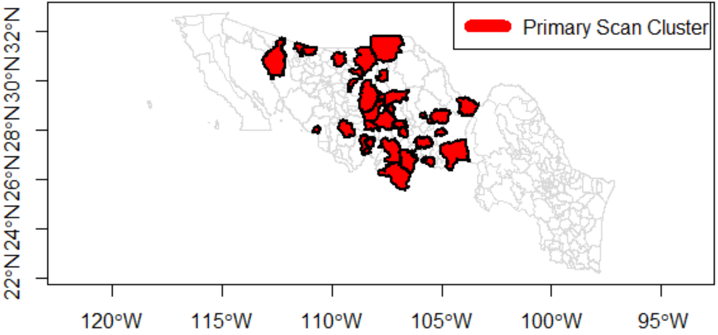


Figure 4: Scan Test

Scan Test (Kulldorff) Municipalities

Border Municipalities	Nearby Municipalities
Manuel Benavides	Ascensión
Matamoros	Empalme
Nogales	Caborca
Janos	Sáric
	Bacerac
	Bacadéhuachi

Table 3: Primary Kulldorff scan cluster municipalities located on or near the U.S.–Mexico border.

- As we see, there are only around 4 border municipalities and 10 total municipalities where we see some sort of clustering. These are pretty much scattered through the central part of Northern Mexico.

Adjusted Moran's I

- Global test

Ho Our measure of global spatial autocorrelation is zero

Ha Our measure of global spatial autocorrelation is not zero

Surface	**Moran's I**	**p-value**
Observed Clinics	0.16714	0.002
Expected Clinics (CRH)	0.16710	0.001

Table 4: Monte Carlo simulation results of Moran's I for observed and expected clinic counts in Northern Mexico (999 simulations).

Adjusted Moran's I Discussion

- Even though this is the adjusted Moran's I using the expected amount of clinics. When run with "number of clinics", the difference in the statistic was insignificant.
- It is important to interpret this though as a sign that what might be driving the spatial autocorrelation is population density.

Conclusion

- These test do seem to indicate there is clustering of observed clinics in the middle region of Sonora and some part of Chihuahua. As these are consistent through the different tests.
- As we saw there were some clustering near the border, however, the clustering was nowhere near these "medical" hubs for medical tourism like Tijuana, Juarez or Monterrey.
- Limitations this is just considering the CRH or that clinics are proportional to populations, this is clearly not the only thing that drives clinics.
- Thus further analysis would be optimal in order to account for several other factors.

References I

Clement, V., Lemus, S., and Newman, E. (2024).
In search of health: medical tourism at the us–mexico border/lands.
Journal of Tourism and Cultural Change, 22(1):118–136.